CityPulse: Intelligent Parking & Traffic Control Platform

Connecting Drivers, Parking Owners & Authorities for Smarter Cities

CSE499A — Section 15 — Project Group 03

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***Abstract*—Anyone who has driven in Dhaka knows the frustration. You are running late, the road is jammed, and somewhere ahead a single car parked on the roadside has turned everything into a mess. Illegal roadside parking is not just an inconvenience in Bangladesh, it is one of the biggest reasons traffic breaks down the way it does every single day. And the current solution of sending traffic police to stand at every busy intersection in the summer heat or heavy rain is simply not working anymore.**

CityPulse is an intelligent parking and traffic management platform built with this problem in mind. It connects three groups of people in one place: drivers who need parking, space owners who have parking, and city or traffic authorities who need better visibility over what is happening on the roads. Parking owners register their spaces and connect existing CCTV cameras, and the system uses computer vision and machine learning to automatically monitor every slot and know in real time which ones are free and which are taken. Drivers simply open the mobile app, see available parking around them on a map, and navigate there directly without any guessing or unnecessary stops on the road.

The platform is designed with a clear research-first approach. This semester focuses on deeply understanding the problem, designing the system properly, and delivering a working prototype that covers parking detection, real-time availability, and navigation. Alongside this, an initial study of Bangla license plate recognition is conducted to lay the groundwork for future intelligent surveillance capabilities.

CityPulse is not a complete solution to every traffic problem Bangladesh has. But it is a serious and realistic starting point built on solid research and a clear understanding of the local context.

***Index Terms*—smart parking, traffic management, computer vision, parking occupancy detection, urban mobility, Bangladesh, machine learning, real-time systems**

I. Introduction

Urban traffic congestion is a growing crisis in Bangladesh, particularly in Dhaka, one of the most densely populated cities in the world. According to a World Bank report, Dhaka's average traffic speed has dropped to around 7 km/h in recent years, making it one of the slowest cities for road travel globally [**?**]. A significant contributor to this problem is illegal and unorganized roadside parking, where vehicles stop wherever they find space, often blocking entire lanes and triggering chain jams that can last for hours.

The traditional response to this problem has been to deploy traffic police at major intersections. While this provides some relief, it is unsustainable. Officers must work in extreme heat, heavy monsoon rain, and polluted air for long hours. The number of problem points across a city like Dhaka far exceeds the available manpower, and human intervention alone cannot scale to meet the demand.

At the same time, Bangladesh has seen rapid growth in smartphone usage and mobile internet penetration. As of 2024, there are over 180 million mobile subscribers in the country, with a steadily increasing share using smartphones and mobile data [**?**]. This creates a real opportunity to build digital solutions that can reach a wide user base and make a measurable difference in daily urban life.

CityPulse is designed to take advantage of this opportunity. It is a smart, AI-powered platform that brings together drivers, parking space owners, and traffic authorities into a single connected system. By automating parking detection and enabling real-time availability search, CityPulse aims to reduce illegal parking, ease congestion, and give both drivers and authorities better visibility over parking and road usage.

This proposal outlines the motivation, objectives, system design, feature breakdown, and implementation plan for CityPulse, developed as a Senior Design I project.

II. Problem Statement

The key problems that CityPulse is designed to address are:

- **Illegal roadside parking:** Drivers park on roadsides due to a lack of visible, accessible, and organized parking options. This directly causes lane blockages and traffic jams.
- **No real-time parking information:** There is currently no system in Bangladesh that tells a driver where nearby parking is available before they arrive. Drivers circle areas looking for space, adding to congestion.
- **Unsustainable manual traffic enforcement:** Deploying traffic police at every congested point is expensive, physically demanding, and impossible to scale across an entire city.
- **Underutilized private parking:** Many private and organizational parking spaces remain partially empty while nearby roads suffer from illegal parking, simply because drivers do not know these spaces exist or are available.

III. Related Work

Several parking and traffic management systems have been proposed and deployed globally, though very few address the Bangladesh context.

Smart parking platforms such as ParkWhiz and SpotHero allow users to find and book parking through mobile apps [?], but rely on manual operator input and established digital infrastructure without any AI-based visual detection.

Computer vision based parking occupancy detection has been widely studied. The PKLot dataset [?] demonstrated that slot occupancy can be detected from fixed overhead cameras with high accuracy, and more recent work applies YOLO-based models for real-time monitoring from CCTV feeds [?].

Within Bangladesh, the Dhaka North City Corporation launched a Smart Parking app in 2023 covering 202 on-street spots in Gulshan [?], and Parking Koi offers a marketplace connecting drivers with private space owners across Dhaka and Chattogram [?]. Both rely on manual availability updates with no AI-based detection, no CCTV integration, and limited geographic coverage.

Bangla script license plate recognition remains under-researched, with very limited publicly available datasets for Bangladeshi plates [?]. Recent work has begun addressing this using YOLO-based detection combined with vision-language OCR [?], [?]. Traffic jam detection using vehicle density estimation and optical flow has also been explored, with recent work specifically targeting Dhaka's heterogeneous traffic conditions [?], [?], [?].

CityPulse builds on these works while addressing the gap left by existing BD systems: automated visual detection using existing CCTV infrastructure, city-wide scalability, and a unified platform for drivers, owners, and authorities.

IV. Objectives

The primary objectives of the CityPulse system are:

1) Build a mobile application for drivers to search, find, and navigate to nearby available parking spaces in real time.
2) Develop an AI-powered computer vision module that processes live CCTV feeds to automatically detect parking slot occupancy.
3) Create a registration and management portal for parking space owners to list their facilities and connect existing cameras.
4) Deliver real-time slot availability updates to the mobile app using a live data pipeline from camera feed to client.
5) Provide a basic authority-facing view showing live parking occupancy statistics across registered spaces.
6) Send push notifications to drivers for parking availability alerts and relevant updates.
7) Conduct an initial study of Bangla license plate recognition approaches, real-time traffic jam detection methods, and vehicle density estimation techniques, and collect preliminary datasets as groundwork for future development.
8) Design and document a complete system architecture that supports scaling to advanced features in subsequent work.

V. System Overview

CityPulse follows a modular, service-oriented architecture that separates concerns across four main layers: the mobile client, the web platform, the backend API, and the AI/ML processing service.

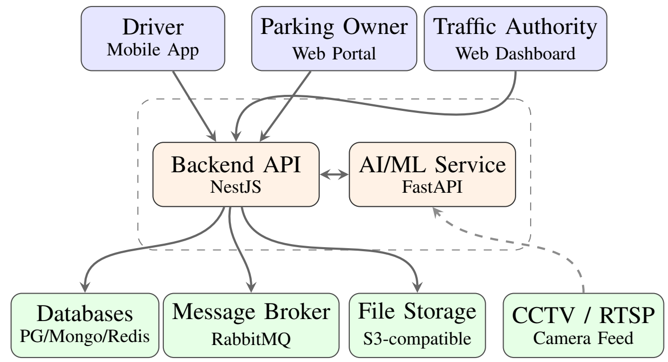

Fig. 1. High-level overview of the CityPulse system.

The system is accessed through three interfaces: a Flutter mobile app for drivers, a Next.js web portal for parking owners, and a Next.js dashboard for traffic authorities. All three communicate with a central NestJS backend via WebSocket, while a dedicated FastAPI service handles AI workloads including live CCTV stream processing and slot occupancy detection.

A. Core Components

Table ?? summarizes the technology stack used across all layers of the CityPulse platform.

TABLE I
CityPulse Technology Stack

Layer	Technology & Purpose
Mobile App	**Flutter** – cross-platform driver app covering parking search, map view, navigation, and push notifications
Web Frontend	**Next.js** – owner registration portal and the authority monitoring dashboard
Backend API	**NestJS** (TypeScript) – authentication, business logic, parking management, and WebSocket real-time updates
AI/ML Service	**FastAPI** (Python) – CCTV stream processing and slot occupancy detection
Relational DB	**PostgreSQL** – stores users, parking spaces, space details, and location data
Document DB	**MongoDB** – stores detection event logs, camera snapshot metadata, and slot status history
Cache	**Redis** – manages real-time slot state cache and pub/sub for live updates
File Storage	**S3-compatible** – object storage for parking photos, CCTV snapshots, and vehicle detection images
Notifications	**Firebase FCM** – push notifications for parking availability alerts and updates
Containerization	**Docker** – service orchestration and deployment via Docker Compose

VI. Features Breakdown

A. User Authentication

CityPulse supports three roles: drivers, parking owners, and authority users, each with a separate registration flow and role-based access control enforced at the API level.

B. Parking Space Registration

Owners register facilities with location, capacity, space type, operating hours, and pricing, and connect existing CCTV cameras via RTSP stream URLs for continuous AI monitoring.

C. Real-Time Slot Availability

The AI service processes live CCTV feeds and detects whether each slot is free or occupied using a computer vision model trained on the PKLot dataset and fine-tuned on locally collected Bangladeshi parking images. Slot status updates are pushed to the mobile app via WebSocket in real time.

D. Parking Search and Map View

Drivers search for parking near their current location. The app displays available spaces on an interactive map with distance, available slot count, price, and space type, sorted by proximity and availability.

E. Navigation to Parking

Once a driver selects a parking space, the app provides turn-by-turn navigation to the entrance using the Google Maps Directions API or OpenRouteService.

VII. Implementation Plan

Each sprint follows an explore-first, then-build approach. Figure **??** shows the planned timeline.

Sprint 1 (Weeks 1–2)

Literature Review & Problem Analysis

Explore:

- Study existing smart parking systems and urban traffic management solutions.
- Review available parking slot detection datasets and benchmark studies.
- Field visit to Dhaka parking lots to observe real camera setups and conditions.
- Study legal and regulatory framework for surveillance and vehicle data in Bangladesh.
- Document findings and identify knowledge gaps.

Build: None — pure study and field work sprint.

Sprint 2 (Weeks 3–4)

System Design & Architecture

Explore:

- Geospatial data storage and proximity query approaches.
- Real-time communication patterns for live data delivery to clients.
- Video stream ingestion and frame processing strategies.

Build:

- Finalize system architecture and data flow diagrams.
- Design full database schema and define all API contracts.
- Prepare UI/UX wireframes for all three client interfaces.
- Set up containerized development environment.

Sprint 3 (Weeks 5–6)

Authentication & Owner Registration

Explore:

- Role-based access control and token-based authentication patterns.
- Multi-role user management flows for web and mobile applications.

Build:

- Role-based authentication for driver, owner, and authority roles.
- Parking space registration with location, capacity, photos, and camera connection.
- Owner portal UI and cloud file storage integration.

Sprint 4 (Weeks 7–8)

Computer Vision & Slot Detection

Explore:

- Object detection and image classification approaches for parking occupancy.
- Fixed-camera computer vision challenges — lighting, angle, occlusion.
- Model optimization and lightweight inference for edge deployment.
- Collect and annotate local parking images from Dhaka.

Build:

- Train and evaluate a slot occupancy detection model on benchmark and local data.
- Build an AI service with live camera stream ingestion and real-time slot status delivery.

Sprint 5 (Weeks 9–10)

Mobile App, Map View & License Plate Recognition Study

Explore:

- Cross-platform mobile map integration and location-based search.
- Real-time data delivery to mobile clients and offline map strategies.
- Existing Bangla license plate recognition approaches and their limitations.
- Available Bangladeshi plate datasets and annotation strategies.
- Collect 100–200 sample Bangladeshi plate images for initial study.

Build:

- Mobile app with interactive map and nearby parking search by location.
- Real-time slot availability display with live updates.
- Basic navigation to selected parking entrance.
- Basic plate localization prototype — detect plate region from a static image.

Sprint 6 (Weeks 11–12)

Integration, Testing & Final Demo

Explore:

- End-to-end testing strategies for distributed systems.

- Performance and load testing for real-time data pipelines.
- Real-time traffic jam detection approaches — vehicle density estimation, optical flow analysis, and video analytics for jam-prone intersections.

Build:

- Integrate all components into a working end-to-end prototype.
- Basic authority view showing live parking occupancy.
- System testing, bug fixes, and demo preparation.

A. Project Timeline

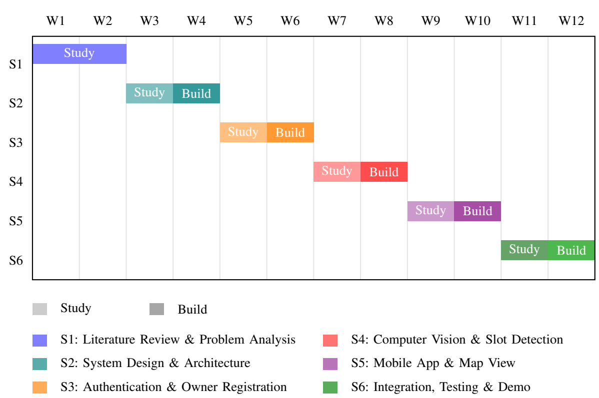

Fig. 2. CityPulse project timeline across 12 weeks.

VIII. Risks and Mitigation

- **CCTV quality & angle variation:** Fine-tune model on local images with manual slot polygon annotation.
- **Bangla plate dataset scarcity:** Initial dataset collection in Sprint 5 with annotation via Roboflow and data augmentation.
- **Real-time performance:** Process frames at 2–3 second intervals with lightweight optimized inference.
- **Owner onboarding:** Simple registration flow with a clear value proposition for parking owners.
- **Privacy and ethics:** Data retention policies, anonymization, and Digital Security Act compliance.

IX. Future Work

The system developed in this phase is intended to serve as a foundation for a more complete intelligent platform. Potential directions for further development include Bangla license plate recognition with a custom annotated dataset, real-time traffic jam detection at road checkpoints, slot booking and reservation, demand-based dynamic pricing, predictive parking availability from historical occupancy data, AI-based vehicle movement analysis, and possible integration with traffic and law enforcement authorities. These areas could build directly on the research and architecture established here.

X. Conclusion

CityPulse is a practical, AI-driven response to one of Bangladesh's most persistent urban problems. By connecting drivers, parking space owners, and traffic authorities in a single intelligent platform, it addresses illegal roadside parking and traffic congestion at a systemic level rather than relying on manual enforcement. The system combines real-time computer vision, mobile-first user experience, and live data delivery into a working prototype that is both technically grounded and locally relevant.

The explore-first development approach ensures that every component is built on a solid research foundation. The project also lays the groundwork for future work including Bangla license plate recognition, predictive parking availability, and smart city surveillance integration.

CityPulse will not solve every traffic problem in Bangladesh overnight. But it demonstrates that with the right combination of technology, thoughtful design, and a clear understanding of the local context, meaningful progress is possible. It is a starting point worth building on.

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